

Mechanics of a Single Particle

Classical Mechanics – Lecture Notes

1 Conservation of Linear Momentum

The net force acting on a particle equals the time rate of change of its linear momentum (Newton's second law):

$$\vec{F} = \frac{d\vec{p}}{dt}, \quad \vec{p} = m\vec{v}. \quad (1)$$

where $\vec{p}$ is the linear momentum. Suppose the net force on the particle vanishes,

$$\vec{F} = \vec{0}. \quad (2)$$

Then equation (1) reduces to

$$\frac{d\vec{p}}{dt} = \vec{0}, \quad (3)$$

which yields

$$\vec{p} = m\vec{v} = \text{constant}. \quad (4)$$

Because both the magnitude and the direction of $\vec{p}$ remain fixed, the particle continues to move in a straight line at constant speed. This is precisely the content of the law of inertia expressed in the language of momentum.

2 Angular Momentum and Torque

Angular momentum is the rotational analogue of linear momentum. Relative to a chosen origin O it is defined by the vector product

$$\vec{\ell} = \vec{r} \times \vec{p}. \quad (5)$$

Unlike linear momentum, the value of $\vec{\ell}$ depends on the location of the origin.

2.1 Derivation of the torque equation

We begin with the definition (5) and differentiate both sides with respect to time:

$$\vec{\tau} \equiv \frac{d\vec{\ell}}{dt} = \frac{d}{dt}(\vec{r} \times \vec{p}). \quad (6)$$

Apply the product rule for the cross product:

$$\frac{d}{dt}(\vec{r} \times \vec{p}) = \dot{\vec{r}} \times \vec{p} + \vec{r} \times \dot{\vec{p}}. \quad (7)$$

The first term is

$$\dot{\vec{r}} \times \vec{p} = \vec{v} \times (m\vec{v}) = m(\vec{v} \times \vec{v}). \quad (8)$$

The cross product of any vector with itself is the zero vector, so

$$\vec{v} \times \vec{v} = \vec{0} \quad \Rightarrow \quad \dot{\vec{r}} \times \vec{p} = \vec{0}. \quad (9)$$

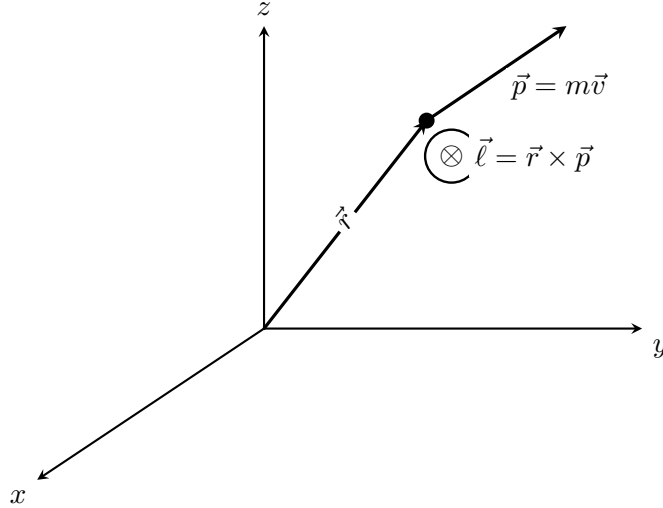


Figure 1: Geometrical representation of the position vector $\vec{r}$, momentum $\vec{p} = m\vec{v}$, and angular momentum $\vec{\ell} = \vec{r} \times \vec{p}$.

Newton's second law supplies the identification of net force: $\dot{\vec{p}} = \vec{F}$. Therefore the second term becomes

$$\vec{r} \times \dot{\vec{p}} = \vec{r} \times \vec{F}. \quad (10)$$

Collecting these results we obtain the fundamental relation

$$\vec{\tau} = \vec{r} \times \vec{F}. \quad (11)$$

Whenever the net torque about a chosen origin vanishes,

$$\vec{\tau} = \vec{0}, \quad (12)$$

equation (6) implies

$$\frac{d\vec{\ell}}{dt} = \vec{0} \quad \Rightarrow \quad \vec{\ell} = \text{constant}. \quad (13)$$

Angular momentum about that origin is therefore conserved.

Central forces and planetary motion

A force is called central when it always points along the line that joins the particle to a fixed centre. For a planet of mass m orbiting the Sun of mass M the gravitational force is

$$\vec{F} = -G \frac{Mm}{r^2} \hat{r}. \quad (14)$$

Taking the origin at the centre of the Sun we have $\vec{F} \parallel \vec{r}$, and therefore

$$\vec{\tau} = \vec{r} \times \vec{F} = \vec{0}. \quad (15)$$

Equation (6) then yields

$$\frac{d\vec{\ell}}{dt} = \vec{0} \quad \Rightarrow \quad \vec{\ell} = \text{constant}. \quad (16)$$

Because $\vec{\ell}$ is constant both in magnitude and in direction, the two vectors $\vec{r}$ and $\vec{p}$ remain forever in the fixed plane perpendicular to $\vec{\ell}$. The planet's orbit is consequently confined to a single plane that contains the Sun, and the original three-dimensional problem reduces to a two-dimensional one.

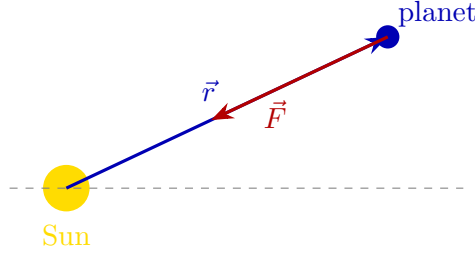


Figure 2: Central gravitational force on a planet. The torque about the Sun vanishes, so angular momentum is conserved.

3 Kinetic Energy and the Work–Energy Theorem

The kinetic energy of a particle of mass m that moves with velocity $\vec{v}$ is defined by

$$T = \frac{1}{2}mv^2 = \frac{1}{2}m(\vec{v} \cdot \vec{v}). \quad (17)$$

Differentiate both sides with respect to time, treating m as constant:

$$\frac{dT}{dt} = \frac{1}{2}m(\dot{\vec{v}} \cdot \vec{v} + \vec{v} \cdot \dot{\vec{v}}) = \frac{1}{2}m(2\vec{v} \cdot \dot{\vec{v}}) = m\vec{v} \cdot \vec{a}. \quad (18)$$

Using Newton’s second law $\vec{F} = m\vec{a}$

$$\frac{dT}{dt} = \vec{F} \cdot \vec{v} = \vec{F} \cdot \frac{d\vec{r}}{dt}. \quad (19)$$

Multiplying through by the infinitesimal time interval dt produces the differential form of the work–energy theorem:

$$\boxed{dT = \vec{F} \cdot d\vec{r}}. \quad (20)$$

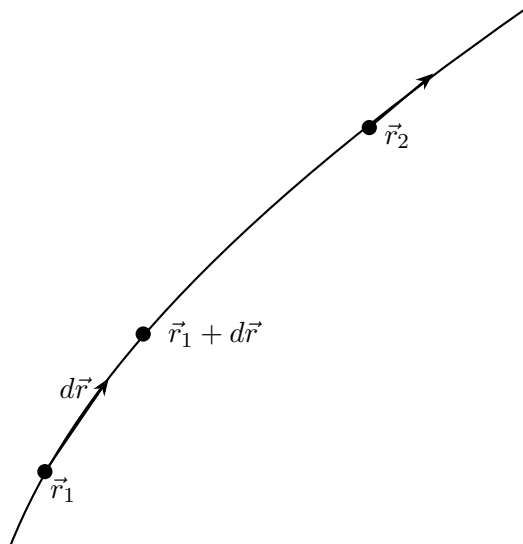


Figure 3: Motion of a particle along a curved trajectory. The particle changes position from $\vec{r}_1$ to $\vec{r}_1 + d\vec{r}$, where $d\vec{r}$ denotes the infinitesimal displacement. A later position is represented by $\vec{r}_2$.

Now integrate along an arbitrary path that begins at point 1 and ends at point 2:

$$\int_{T_1}^{T_2} dT = \int_1^2 \vec{F} \cdot d\vec{r}. \quad (21)$$

The left-hand side evaluates immediately to $T_2 - T_1$. The right-hand side is, by definition, the work $W(1 \rightarrow 2)$ performed by the force along the path. Hence

$$T_2 - T_1 = W(1 \rightarrow 2) = \int_1^2 \vec{F} \cdot d\vec{r}. \quad (22)$$

The sign of the infinitesimal work $\vec{F} \cdot d\vec{r}$ determines how the kinetic energy changes:

- $dT > 0$ when $\vec{F}$ and $d\vec{r}$ point in the same direction,
- $dT = 0$ when $\vec{F}$ is perpendicular to $d\vec{r}$,
- $dT < 0$ when $\vec{F}$ and $d\vec{r}$ point in opposite directions.

4 Potential Energy and Conservative Forces

Not every force has an associated potential energy. Those forces that do are called *conservative forces*. A force $\vec{F}$ is conservative if and only if it satisfies the following two conditions:

- $\vec{F}$ depends only on the particle's position, i.e. $\vec{F} = \vec{F}(\vec{r})$ (it is independent of velocity and of time);
- the work $W(1 \rightarrow 2)$ done by the force is independent of the particular path that joins the two points, i.e., the work done depends solely on the endpoints.

Classic examples of conservative forces are the Newtonian gravitational force

$$\vec{F} = -G \frac{m_1 m_2}{r^2} \hat{r} \quad (23)$$

and the electrostatic Coulomb force

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}. \quad (24)$$

Forces that fail at least one of the two conditions (air resistance, kinetic friction, the magnetic force on a moving charge, and forces arising from explicitly time-dependent fields) are non-conservative.

4.1 Definition of potential energy

To construct the potential energy from a given conservative force, we first select a convenient reference point $\vec{r}_0$ at which we set

$$U(\vec{r}_0) = 0. \quad (25)$$

The potential energy at an arbitrary point $\vec{r}$ is then defined to be the negative of the work required to bring the particle from the reference point to $\vec{r}$:

$$U(\vec{r}) = -W(\vec{r}_0 \rightarrow \vec{r}) = - \int_{\vec{r}_0}^{\vec{r}} \vec{F}(\vec{r}') \cdot d\vec{r}'. \quad (26)$$

Because the work is path-independent, the line integral is uniquely defined; its value depends only on the endpoints.

From the definition, the work performed between any two points (say from $\vec{r}_1$ to $\vec{r}_2$) equals the decrease in potential energy:

$$\begin{aligned} W(1 \rightarrow 2) &= W(\vec{r}_0 \rightarrow \vec{r}_2) - W(\vec{r}_0 \rightarrow \vec{r}_1) \\ &= -(U(\vec{r}_2) - U(\vec{r}_1)) = -\Delta U. \end{aligned} \quad (27)$$

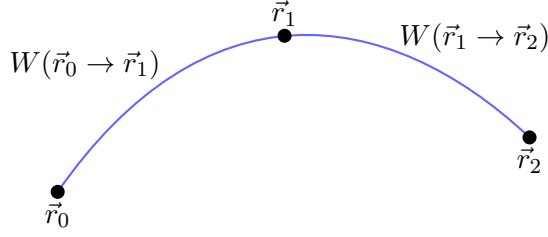


Figure 4: Path-independent work used to define potential energy. The reference point $\vec{r}_0$ is chosen so that $U(\vec{r}_0) = 0$.

Combine this identity with the work–energy theorem (22):

$$\Delta T = -\Delta U \quad \Rightarrow \quad \Delta(T + U) = 0. \quad (28)$$

Consequently the mechanical energy

$$E = T + U \quad (29)$$

Thus, if the force on a particle is conservative, then the particles mechanical energy never changes, i.e., particle’s energy is conserved.

4.2 Several conservative forces

Suppose the particle feels several conservative forces $\vec{F}_1, \dots, \vec{F}_n$ with corresponding potentials $U_1, \dots, U_n$. The total work is the sum of the individual works,

$$W_{\text{tot}} = W_1 + \dots + W_n = -\Delta U_1 - \dots - \Delta U_n. \quad (30)$$

The work–energy theorem then reads

$$\Delta T = -\Delta(U_1 + \dots + U_n), \quad (31)$$

or

$$\Delta(T + U_1 + \dots + U_n) = 0. \quad (32)$$

The total mechanical energy is therefore still conserved.

4.3 Mixture of conservative and non-conservative forces

When both conservative and non-conservative forces act, the work–energy theorem becomes

$$\Delta T = W_c + W_{\text{nc}} = -\Delta U_c + W_{\text{nc}}, \quad (33)$$

which rearranges to

$$\Delta(T + U_c) = W_{\text{nc}}. \quad (34)$$

Mechanical energy is no longer conserved. However, the difference in energy equals exactly the work performed by the non-conservative forces. In most cases, work done by the non-conservative forces are negative, $W_{\text{nc}} < 0$, (e.g., kinetic friction works in the direction opposite to the motion) so the particle steadily loses mechanical energy by the amount W_{nc} .

5 Force as the Gradient of Potential Energy

Start from the definition (26) and consider an infinitesimal displacement $d\vec{r}$. The corresponding change in potential energy is

$$dU = -\vec{F} \cdot d\vec{r} = -(F_x dx + F_y dy + F_z dz). \quad (35)$$

On the other hand, if U is a differentiable function of the Cartesian coordinates, its total differential is

$$dU = \frac{\partial U}{\partial x} dx + \frac{\partial U}{\partial y} dy + \frac{\partial U}{\partial z} dz. \quad (36)$$

Equating the two expressions for dU and comparing coefficients of the independent differentials dx , dy and dz yields

$$F_x = -\frac{\partial U}{\partial x}, \quad F_y = -\frac{\partial U}{\partial y}, \quad F_z = -\frac{\partial U}{\partial z}. \quad (37)$$

In vector notation this is simply

$$\vec{F} = -\nabla U, \quad (38)$$

where the gradient operator is defined by

$$\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}. \quad (39)$$

Thus every conservative force is equal to the negative gradient of its potential energy.

5.1 Curl test for conservativeness

Take the curl of both sides of (38):

$$\nabla \times \vec{F} = -\nabla \times (\nabla U). \quad (40)$$

A fundamental identity of vector calculus states that the curl of any gradient vanishes identically,

$$\nabla \times (\nabla U) = \vec{0}, \quad (41)$$

provided the second partial derivatives of U are continuous. Consequently

$$\nabla \times \vec{F} = \vec{0}. \quad (42)$$

Conversely, if a continuously differentiable force field satisfies $\nabla \times \vec{F} = \vec{0}$ throughout a simply connected domain, then there exists a scalar potential U such that $\vec{F} = -\nabla U$. The vanishing of the curl is therefore a necessary and sufficient condition for a force to be conservative.

We can also show this as,

$$\vec{\nabla} \times \vec{F} = -(\vec{\nabla} \times \vec{\nabla} U) \quad (43)$$

$$= - \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial U}{\partial x} & \frac{\partial U}{\partial y} & \frac{\partial U}{\partial z} \end{vmatrix} \quad (44)$$

$$= - \left\{ \hat{i} \left[\frac{\partial^2 U}{\partial y \partial z} - \frac{\partial^2 U}{\partial z \partial y} \right] + \hat{j} \left[\frac{\partial^2 U}{\partial z \partial x} - \frac{\partial^2 U}{\partial x \partial z} \right] + \hat{k} \left[\frac{\partial^2 U}{\partial x \partial y} - \frac{\partial^2 U}{\partial y \partial x} \right] \right\} \quad (45)$$

$$= 0. \quad (46)$$

So, given a force $\vec{F}(\vec{r})$, a necessary and sufficient condition for the force to be conservative is

$$\boxed{\vec{\nabla} \times \vec{F} = 0}. \quad (47)$$

6 Time-Dependent Potential Energy

Suppose the potential energy depends explicitly on time as well as on position, $U = U(\vec{r}, t)$. Its total differential then acquires an additional term:

$$dU = \frac{\partial U}{\partial x} dx + \frac{\partial U}{\partial y} dy + \frac{\partial U}{\partial z} dz + \frac{\partial U}{\partial t} dt \quad (48)$$

$$= \nabla U \cdot d\vec{r} + \frac{\partial U}{\partial t} dt. \quad (49)$$

Because $\vec{F} = -\nabla U$ we may rewrite the expression as

$$dU = -\vec{F} \cdot d\vec{r} + \frac{\partial U}{\partial t} dt. \quad (50)$$

From the work–energy theorem we still have $dT = \vec{F} \cdot d\vec{r}$. Adding the two differentials yields

$$dT + dU = \frac{\partial U}{\partial t} dt, \quad (51)$$

or

$$dE = \frac{\partial U}{\partial t} dt \quad \Rightarrow \quad \frac{dE}{dt} = \frac{\partial U}{\partial t}. \quad (52)$$

Mechanical energy is no longer conserved; its instantaneous rate of change equals the explicit partial derivative of the potential with respect to time.

7 Summary

- Linear momentum $\vec{p}$ is conserved when the net force vanishes.
- Angular momentum $\vec{\ell} = \vec{r} \times \vec{p}$ is conserved when the net torque vanishes. Central forces automatically produce zero torque about the force centre.
- The work–energy theorem states that the change in kinetic energy equals the work done by the net force.
- A force is conservative when it depends only on position and its work is path-independent. In that case a potential energy U exists, $\vec{F} = -\nabla U$, and the mechanical energy $E = T + U$ is conserved.
- The necessary and sufficient condition for a smooth force field to be conservative is $\nabla \times \vec{F} = \vec{0}$.
- When non-conservative forces are present the change in mechanical energy equals the work they perform.
- If the potential depends explicitly on time, the rate of change of mechanical energy equals $\partial U / \partial t$.